Document Identifier: **PMC8325782**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: All experiments were carried out as per protocol approved by the Institutional Animal Care and Use Committee, Iowa State University (IACUC 18-059060).
Euthanasia Agents: At the end of the study, all animals were euthanized with pentobarbital sodium (100 mgkg, i.p.). 2.2.
Inclusion and Exclusion Criteria
Electric and behavior artifacts were detected and excluded from the analysis (Puttachary et al. , 2015 Puttachary et al. , 2016b) .
Attrition
not detected.
Sex as a biological variable
not detected.
Subject Demographics
Age: Animal source and ethics statement We used adult male Sprague Dawley rats (8 weeks old) from Charles River Laboratory (MA, USA) in this study.
Randomization
The animals were randomized and coded to blind the experimental groups .
Blinding
The animals were randomized and coded to blind the experimental groups .
Power Analysis
not detected.
Replication
All reactions were performed in triplicate.

Number: All reactions were performed in triplicate.
Cell Line Authentication
Authentication: We took measures to minimize other variables and biases e.g., i) direct observation of SE and secondary video analysis by a different person who was blind to the protocol to determine the initial SE severity, and ii) all SRS were validated against the synchronized video, power spectrum, and activity counts as described previously (Puttachary et al., 2015).

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Antibodies were purchased from different sources: glial fibrillary acidic protein (GFAP , a marker for astrocytes , mouse and rabbit monopolyclonal , 1:400) from Sigma and Novus Biologicals , USA ionized calcium-binding adaptor molecule 1 (IBA1 , a marker for microglia , goat polyclonal , 1:400) , 3-nitrotyrosine (3-NT , mouse monoclonal , 1:1400) , 4-hydroxynonenal (4-HNE) , and glycosylated glycoprotein (gp91phox , rabbit polyclonal , 1:1000) from Abcam , USA Fyn (mouse monoclonal , 1:200 for IHC and 1:800 for WB) , protein kinase C delta (PKCd , rabbit polyclonal , 1:300 for IHC and 1:1000 for WB) , phosphorylated PKCd (pPKCd-Ty311	glial fibrillary acidic protein		
	GFAP		
	ionized calcium-binding adaptor molecule 1		
	IBA1		
	3-nitrotyrosine (3-NT ,		
	4-hydroxynonenal (4-HNE) , and glycosylated glycoprotein (gp91phox		
	protein kinase C delta (PKCd ,		
, goat polyclonal , 1:1000) were all from Santa Cruz Biotechnology , USA neuronal nuclei marker (NeuN , rabbit polyclonal , 1:500) and phosphorylated SrcFyn specific antibody (pSrc-416 , rabbit monoclonal , 1:1000) were from Millipore and Cell Signaling , USA Fluoro-Jade B was purchased from Histochem Inc . , USA .	NeuN		
	pSrc-416		
All secondary antibodies for IHC (CY3 conjugated 1:200	CY3		
Experimental Models: Organisms/Strains			
Animal source and ethics statement We used adult male Sprague Dawley rats (8 weeks old) from Charles River Laboratory (MA, USA) in this study.	Sprague Dawley		Suggestion: RRID:RGD_70508(link)
Recombinant DNA			
The other half was used for whole tissue lysate to determine pSrc-416 , pPKCd-Tyr311 , and nitroxidative stress markers .	pPKCd-Tyr311		
Software and Algorithms			
Antibodies were purchased from different sources: glial fibrillary acidic protein (GFAP , a marker for astrocytes , mouse and rabbit monopolyclonal , 1:400) from Sigma	Sigma		Suggestion: sigma-tau (RRID:SCR_000488)(link)

and Novus Biologicals , USA ionized calcium-binding adaptor molecule 1 (IBA1 , a marker for microglia , goat polyclonal , 1:400) , 3-nitrotyrosine (3-NT , mouse monoclonal , 1:1400) , 4-hydroxynonenal (4-HNE) , and glycosylated glycoprotein (gp91phox , rabbit polyclonal , 1:1000) from Abcam , USA Fyn (mouse monoclonal , 1:200 for IHC and 1:800 for WB) , protein kinase C delta (PKCd , rabbit polyclonal , 1:300 for IHC and 1:1000 for WB) , phosphorylated PKCd (pPKCd-Ty311			
Quantification of epileptiform spikes and spontaneously recurring seizures (SRS) All the epileptiform spikes , including spike clusters and interictal spikes , CS , and NCS were quantified based on the spike characteristics and spike frequency using the NeuroScore 3.2.0 as described in our previous publications (Puttachary et al. , 2015 and Puttachary et al. , 2016b Sharma et al. , 2018a Tse et al. , 2014) .	NeuroScore		
The following day , after couple of washes with PBS-T , membranes were incubated with IR-680 or IR-800 dyes (1:5000-1:10,000 , Li-Cor) , and analyzed using Odyssey IR imaging system and quantified using ImageJ software.	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
NanoDrop spectrophotometer was used to quantify RNA , and 1 mg of RNA was reversed transcribed to cDNA using High Capacity cDNA Reverse Transcription Kit (catalog 4368814 , Applied Biosystems , CA , USA)	NanoDrop		
Prism 8 (GraphPad Software, LLC) was used for data analysis. 2.14.	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Plates were read at 540 nm using Synergy 2 Multi-mode Microplate Reader (BioTek Instruments, USA) and the averages were compared between the groups using ANOVA. 2.12.	ANOVA
Statistical analyses For statistical analyses, the Gaussian distribution of data was tested with the Shapiro-Wilk and Kolmogorov-Smirnov tests.	Shapiro-Wilk
	Kolmogorov-Smirnov tests
Unpaired t-tests were used for parametric comparisons, and the Mann-Whitney U (MWU) test was used for unpaired non-parametric data.	t-tests
	Mann-Whitney U (MWU) test
One-way repeated analysis of variance (ANOVA) was used for multiple comparisons of parametric data with Tukeys post-hoc analysis.	analysis of variance
	Tukeys
To analyze gliosis and neurodegeneration in different brain regions of the same animal, we used repeated measures (RM) two-way ANOVA (Figs. 5B-D, 7B-D, and 11B) with Tukeys post-hoc test.	Tukeys post-hoc test
For seizure severity and spike analysis, two-way ANOVA (or mixed-effects model) with the Geisser-Greenhouse correction was used for multiple comparisons with Sidak post-hoc test, which gives multiplicity adjusted p-value for each comparison.	Sidak post-hoc test

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq